Risk-Aware Portfolio for the Serendib University Endowment Fund

QuantEdge 1.0 Final Challenge · technical report and investment memo

Team Gmora

USD 10M · seven ETFs · decision date 2026-10-26 · market data to 2026-10-23

1. Executive summary
2. Client mandate and Investment Policy Statement
3. Data
4. Portfolio construction
5. Risk measurement
6. Model validation
7. Champion–challenger and final test
8. Stress testing
9. Risk budget and recommendation
10. Assumptions and limitations
11. References
12. AI disclosure

1. Executive summary

Decision of 2026-10-26: **ACCEPT** at 35% equity

The Serendib University Endowment Fund asks for capital preservation, income and modest real growth, and it values avoiding large losses above maximising gains. We recommend that the committee **accepts the mandate as proposed**: invest 35% in equities and 65% in bonds, with the objective of US inflation + 3.5% a year. The result comes from a rule we fixed before looking at the final test data, and it is a close call that we state openly.

	Weight	USD
AGG	65.0%	6,499,999
SPY	15.6%	1,557,349
XLE	10.0%	1,000,000
EFA	9.4%	942,649

	Result
Expected return a year	6.04%
Required return	5.94%
Worst crisis replay	23.9% (GFC 2008)
Our scenario	21.0%
99% ES, full cycle	USD 208,295
Income a year	USD 326,787
One-shot test	9 breaches vs 9.6 expected

- **Return:** 6.04% expected after costs against 5.94% required, a margin of about USD 10,700 a year.
- **Crisis risk:** a repeat of 2008 costs 23.9%, inside the 25% budget. Our own stagflation scenario costs 21.0%.
- **Daily risk:** about USD 208,000 on the worst 1 day in 100 through the cycle, against a limit of about USD 226,000.
- **Confidence in the model:** GJR conditional EVT passed a one-shot test on 2023–2026, with 9 breaches against 9.6 expected.

The margin is thin. The verdict rests on an assumed 4.5% equity premium and on AGG's interest being free of US withholding tax. It holds in 7 of 18 assumption combinations, and the chance of meeting the objective over ten years is about even. We ask the committee to treat the objective as a target, not a promise, and to take tax advice before investing.

2. Client mandate and Investment Policy Statement

Client and horizon

The fund is a perpetual USD 10M endowment supporting scholarships, research and faculty development. A perpetual fund can wait out a market fall, but it cannot stop paying scholarships during one. We therefore judge the return objective over rolling 10-year periods, monitor risk daily and over 10 days, and budget risk per crisis episode.

Return objective

The fund must pay for its spending, hold its real value and grow a little. We set spending at 3.0% of a 12-quarter average value (range considered 2.5–3.5%), add 0.5% real growth, and use the 10-year breakeven inflation rate at the decision date.

$$\text{Required return} = (1 + 3.0\%) \times (1 + 0.5\%) \times (1 + 2.34\%) - 1 = 5.94\% \text{ a year}$$

Risk constraints

One preference sets the risk limits: the most the committee will lose in any crisis, historical or designed. We set this crisis budget at 25% (range considered 20–30%). The frontier point whose crisis loss equals 25% then fixes the volatility limit (8.29%) and the 99% ES limit (2.26%), so the limits cannot contradict one another. An earlier draft (volatility 8%, crisis loss 20%, ES 1.5%) was replaced because its three limits implied three different equity shares.

	Maximum	Portfolio	Source
Crisis budget	25.00%	23.90%	Team's stated tolerance; sets the two below
Volatility, full cycle	8.29%	7.67%	Frontier point at the crisis budget
99% 1-day ES, full cycle	2.26%	2.08%	Frontier point at the crisis budget
Equity	50.00%	35.00%	Conservative mandate
AGG	65.00%	65.00%	Leaves 35% for equity and gold
XLE	10.00%	10.00%	Sector concentration
Largest share of ES, any ETF	40.00%	33.15%	Position sizing (booklet 8.6)

Table 1. Every limit against the recommended portfolio. AGG and XLE sit at their caps.

The limits were set with knowledge of results up to 2022. The protection against fitting them to that history is the one-shot test that followed.

3. Data

Sources

All data come from the approved sources. Prices are Yahoo Finance adjusted closes for the seven ETFs, through `yfinance`. Rates, inflation expectations and stress factors are FRED series (10-year and 2-year Treasury, 10-year TIPS, breakeven inflation, 3-month T-bill, Baa spread, oil, dollar index, VIX, CPI). Sri Lankan inflation is World Bank data, used for context. The equity premium is a disclosed assumption, not data.

The raw download is frozen in `data/raw/` with a SHA-256 hash for every file, so the judges can rebuild every number from the repository.

Cleaning

- **Period:** 18 November 2004, when GLD began trading, to 26 October 2026: 5,518 trading days.
- **Calendars:** all seven ETFs trade on US exchanges, so the inner join on trading dates dropped 0 days. FRED series have their own holidays and were carried forward by at most 3 business days.
- **Adjusted prices:** used throughout, so dividends and splits do not distort returns. We verified this against close prices plus dividends.
- **Log and simple returns:** log returns for statistics on one series; simple returns wherever assets are combined, because only simple returns add across assets.
- **Risk-free rate:** the 3-month T-bill, lagged one day, so no estimate uses information from the same day.
- **Partition:** development 2004–2018, validation 2019–2022, final test 2023–2026. Every selection used data to 2022 only.

Quality flags

Suspect points are flagged and kept, never deleted. Every one matches a real event, such as the AGG fall of 10 October 2008 and the COVID crash, and none is a data error. Deleting them would remove the crises the fund must survive.

	Points	First	Last
jump	24	2007-02-27	2026-01-30
large_negative	1	2020-03-09	2020-03-09

Table 2. 25 flagged points by rule. All are kept and each was checked against a market event.

Statistical properties

ADF and KPSS agree: log prices have a unit root and log returns are stationary. Returns have fat tails and volatility clustering in every series (Jarque-Bera and ARCH-LM p ≈ 0). This is why a normal model is not enough, and why GARCH and EVT are in the study. AGG has the fattest tails of all, because bonds are usually calm and occasionally are not.

	Volatility	Skew	Excess kurtosis	Jarque-Bera p	ARCH-LM p	ADF p, log price	ADF p, log return	KPSS p, log return
SPY	18.83%	-0.30	14.8	0.000	0.000	0.989	0.000	0.10
EFA	20.92%	-0.35	13.7	0.000	0.000	0.834	0.000	0.10
EEM	27.49%	0.02	15.6	0.000	0.000	0.193	0.000	0.10
AGG	5.21%	-2.09	55.3	0.000	0.000	0.365	0.000	0.10
GLD	18.30%	-0.45	6.6	0.000	0.000	0.769	0.000	0.10
XLE	29.78%	-0.71	12.9	0.000	0.000	0.687	0.000	0.10
XLK	22.58%	-0.16	9.4	0.000	0.000	0.996	0.000	0.10

Table 3. Daily log returns, 2004-11-18 to the snapshot. KPSS p-values are bounded at 0.10.

The sample excludes the 2000–02 technology crash and the 1970s inflation, a limitation we return to in section 10.

4. Portfolio construction

Method

We start from Markowitz mean-variance and CAPM, but on these assets textbook optimisation pushes the portfolio into corners, because small errors in expected returns become large positions. We therefore separate the two jobs. The frontier minimises variance at each equity share under the caps, using a Ledoit-Wolf covariance matrix on 756 days. Expected returns are then used only to score the points.

$$\min w^{T}\Sigma w \text{ subject to } \Sigma\, w_i = 1,\ w_i \geq 0,\ \Sigma_\text{equity}\ w_i = e,\ w_i \leq \text{cap}_i$$

Expected returns are built from yields on the decision date, not from past returns. AGG earns the 10-year yield plus 0.5%. Equities earn the yield plus beta times a 4.5% premium (CAPM). Gold earns breakeven inflation. Fees, withholding tax, trading costs and the variance drag $\frac{1}{2}w^{T}\Sigma w$ are deducted.

Selection rule

The locked rule picks the lowest-risk frontier point that meets the objective inside every limit. It gives **ACCEPT at 35% equity**: SPY 15.6%, EFA 9.4%, AGG 65.0%, XLE 10.0%.

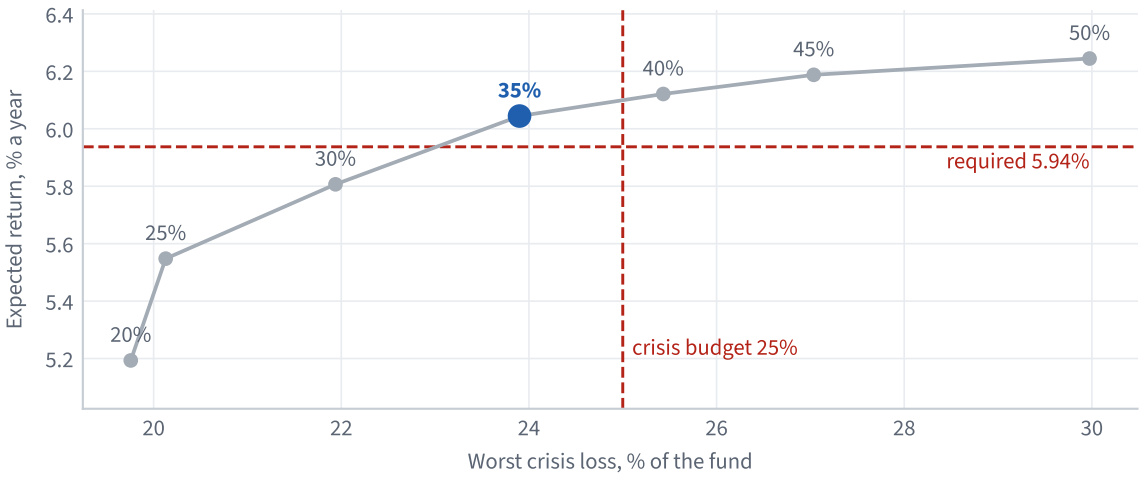


Figure 1. Equity share on each point. Only 35% is above the required return and left of the crisis budget.

	Return	Volatility	99% ES	Crisis loss	Income	Limits	Objective	Caps binding
20%	5.19%	6.32%	1.52%	19.76%	2.96%	yes	no	AGG, GLD
25%	5.55%	6.46%	1.63%	20.13%	3.00%	yes	no	AGG, XLE
30%	5.81%	6.96%	1.84%	21.94%	3.13%	yes	no	AGG, XLE
35%	6.04%	7.67%	2.08%	23.90%	3.27%	yes	yes	AGG, XLE
40%	6.12%	8.53%	2.33%	25.43%	3.16%	no	yes	XLE
45%	6.19%	9.43%	2.57%	27.04%	3.05%	no	yes	XLE
50%	6.24%	10.35%	2.83%	29.98%	2.95%	no	yes	XLE

Table 4. The frontier on the decision date, by equity share. Volatility and ES are through the cycle (2004–2018).

The trade-off

Moving from 30% to 35% equity adds 0.23 points of return, about USD 23,800 a year, for 2.0 points more crisis loss. Moving from 35% to 40% adds only 0.08 points, about USD 7,700 a year, for 1.5 points more crisis loss, and breaks the budget. The gain shrinks because of what equity replaces. Up to 35% it replaces gold, which is assumed to earn only breakeven inflation. Above 35% the gold is gone and it replaces AGG, which yields about 5.5%. So 35% is both the first point that meets the objective and the point where more risk starts to pay much less.

Conventional alternatives

At the decision date, equal weight, 60/40, maximum Sharpe and risk parity all break the crisis budget (losses of 28–42%). Global minimum variance and minimum CVaR stay inside it but are expected to earn only 4.3–4.5%, short of the objective.

History

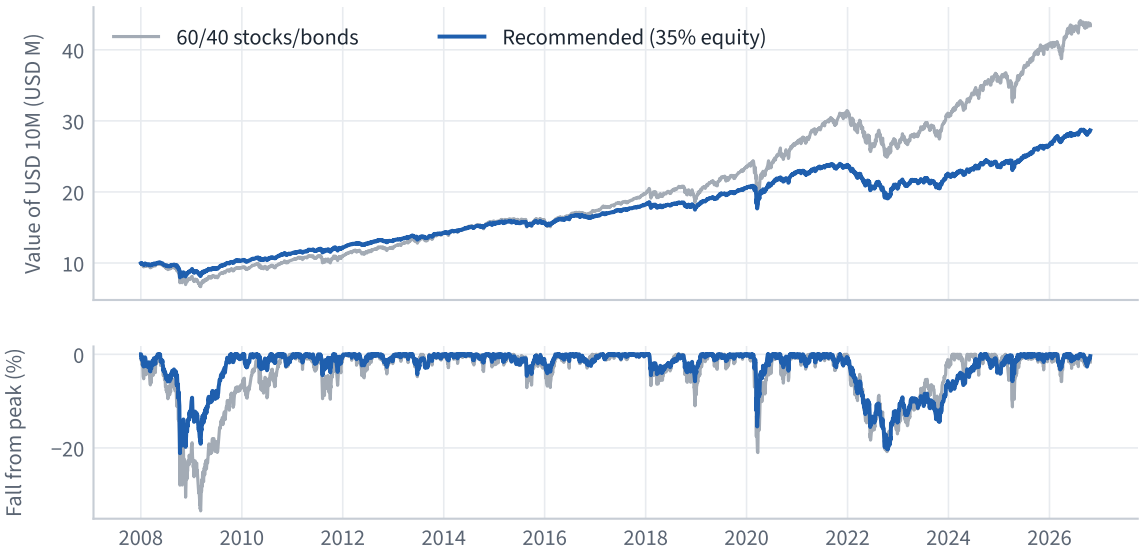


Figure 2. Walk-forward history after 5 bp costs: weights rebuilt each quarter from prices known at the time.

	Recommended, 2008–2026	60/40, 2008–2026	Recommended, 2023–2026	60/40, 2023–2026
Return a year	5.77%	8.13%	9.23%	14.07%
Volatility	7.37%	11.75%	6.40%	9.53%
Sharpe ratio	0.60	0.60	0.71	0.96
Sortino ratio	0.85	0.85	1.01	1.42
Beta to SPY	0.32	0.59	0.31	0.62
Treynor ratio	14.11%	12.10%	14.60%	14.63%
Jensen's alpha	+0.95%	+0.58%	-0.42%	-0.83%
Information ratio vs 60/40	-0.46	–	-0.84	–
Maximum drawdown	21.1%	33.5%	6.6%	11.2%
Calmar ratio	0.27	0.24	1.39	1.25

Table 5. Performance KPIs. 2023–2026 is the period no choice was made on. Beta and alpha are against SPY.

Since 2008 the strategy earned 5.8% a year with a worst fall of 21.1%. The 60/40 earned 8.1% with a worst fall of 33.5%. Their Sharpe ratios are equal at 0.60, so the recommended mix gave up return for less risk rather than for worse risk. Its Treynor ratio and Jensen's alpha are higher because it carries less market risk. The information ratio is negative, as it must be for a conservative mix measured against a riskier benchmark. In 2023–2026, a strong period for equities, the 60/40 was clearly ahead on return and Sharpe. That is the cost of the mandate, and we do not claim otherwise.

5. Risk measurement

Definitions

VaR and ES are positive numbers meaning losses. Losses are computed on point-in-time weights, with simple returns aggregated across assets.

$$\mathrm{VaR}_\alpha = -\inf\{\, x : P(R \le x) \ge 1-\alpha \,\}$$

$$\mathrm{ES}_\alpha = 1/(1-\alpha) \times \int \text{ from } \alpha \text{ to } 1 \text{ of } \mathrm{VaR}_u \, du$$

Three methods, and why they disagree

We compute historical simulation, parametric (normal, Student-t, Cornish-Fisher) and Monte Carlo VaR at 95% and 99%, plus EVT and GARCH-based models, on the same recommended weights.

	VaR 95%	VaR 99%	ES 95%	ES 99%
Historical	53	115	92	169
Normal	61	88	77	101
Student-t	52	100	85	151
Cornish-Fisher	49	212	155	363
Monte Carlo	52	99	84	146
EVT	53	110	94	181
GARCH-FHS	45	83	67	113
GJR-FHS	43	80	64	103
Conditional EVT	44	80	67	110
GJR conditional EVT	42	76	64	101

Table 6. One-day loss in USD thousand on the final weights, each model fitted on 2024-10-25 to 2026-10-23.

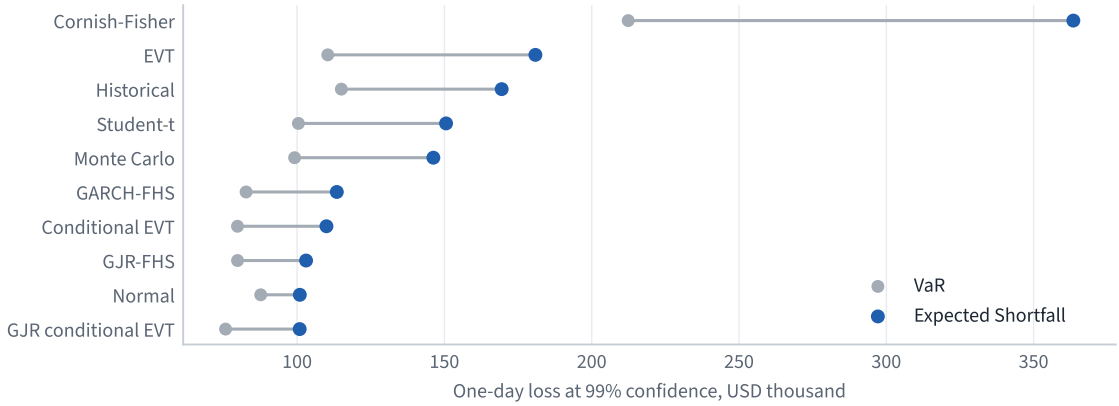


Figure 3. At 99%, how much further ES reaches beyond VaR, by model.

- **At 95% the historical, parametric and Monte Carlo methods nearly agree** (USD 49,000–61,000). This part of the distribution is well observed.
- **At 99% they do not** (USD 88,000–115,000, and USD 212,000 for Cornish-Fisher). Among these methods the normal model is the highest at 95% and the lowest at 99%: tail events inflate its variance, which raises the centre, while its thin tails miss the extremes.
- **Monte Carlo** draws from a fitted Student-t and lands beside the parametric t model, because it shares the same assumption. Simulation adds no information the distribution does not already hold.
- **Volatility-aware models are lower** because they respond to today's calm market, while historical models still remember every bad day in their two-year window.
- **Cornish-Fisher is unreliable here.** Its correction is valid only for mild skew and kurtosis, and these returns are too fat-tailed. We show it for comparison and do not use it.

These figures describe a calm market. Through the cycle, on 2004–2018 data including the 2008 crisis, the average very bad day costs USD 208,000, which is the figure tested against the limit.

Why Expected Shortfall as well

VaR says where a bad day starts and nothing about how bad it gets. It is also not subadditive, so two positions can show more VaR together than apart (Artzner et al., 1999), as a two-bond example in notebook 03 shows. ES averages the whole tail and is coherent. The committee's tail limit is therefore set on ES, with VaR kept because regulators' backtests are built on it.

Extreme Value Theory

We fit a Generalised Pareto distribution to losses above a threshold u (peaks over threshold; Balkema and de Haan, 1974; Pickands, 1975), by maximum likelihood with n observations and N_u exceedances.

$$\mathrm{VaR}_p = u + (\beta / \xi) \times \{ [(n / N_u)(1 - p)]^{(-\xi)} - 1 \}$$

$$\mathrm{ES}_p = (\mathrm{VaR}_p + \beta - \xi u) / (1 - \xi)$$

The threshold was not chosen arbitrarily. Six candidates were compared on parameter stability, Anderson-Darling and Kolmogorov-Smirnov fit, and exceedance counts, and three selection rules were compared. We fixed 90% on the candidate portfolio before the decision date. Repeating the evidence on the final weights shows a weakness we report rather than hide: at 87.5–92.5% the fit is rejected or borderline at 5%, while 95% fits well but gives a heavier tail (ξ of 0.41 against 0.27).

	Exceedances	ξ	s.e.	AD p	KS p	VaR 99%	ES 99%
85.0%	684	0.278	0.047	0.305	0.145	1.44%	2.24%
87.5%	570	0.279	0.051	0.120	0.030	1.44%	2.24%
90.0%	456	0.273	0.055	0.025	0.030	1.44%	2.23%
92.5%	342	0.298	0.066	0.070	0.045	1.43%	2.26%
95.0%	228	0.413	0.093	0.470	0.820	1.39%	2.38%
97.5%	114	0.456	0.150	0.275	0.475	1.37%	2.44%

Table 7. GPD threshold evidence on the final weights, 2004–2022 (no test data). AD and KS are bootstrap goodness-of-fit p-values.

The practical effect is contained. 99% VaR moves only from 1.37% to 1.44% across thresholds, but 99% ES moves from 2.23% to 2.44%, and it would be about 7% higher at the better-fitting 95% threshold. The p-values also assume independent exceedances, which volatility clustering breaks. We read the far tail as a range, not a point. The once-in-a-decade daily loss is USD 162,000 on today's volatility and up to USD 384,000 on the long history. The decision does not rest on this estimate: the ES limit uses the through-the-cycle figure, and the champion applies EVT to GARCH residuals.

6. Model validation

Design

Every model forecasts each day from data up to the day before, using the weights actually held. Forecasts are refitted every 5 days on a 500-day trailing window. We selected on 2019–2022 and tested once on 2023–2026, after a git tag (`methodology-freeze-v1`) had frozen the method.

Tests

Kupiec's proportion-of-failures test asks whether the number of breaches x in T days matches the coverage p.

$$LR_uc = -2 \ln [(1 - p)^(T - x) p^x / ((1 - x/T)^(T - x) (x/T)^x)] \sim \chi^2(1)$$

Christoffersen's independence test asks whether breaches cluster, using the transition counts between breach and no-breach days. Added to Kupiec's statistic it gives conditional coverage, $\chi^2(2)$. **The Basel traffic light** counts 99% breaches in each 250-day window: green up to 4, yellow 5–9, red 10 or more. We add the Acerbi-Székely Z2 test for ES, since VaR tests cannot see ES. Models that pass are ranked by the pinball loss, a proper score for quantiles.

$$L = (y - q)(\tau - 1\{y < q\})$$

Selection on 2019–2022

A model passes if it clears Kupiec and conditional coverage at 5%, has no red rolling window, and has a green Z2. The rule, fixed in advance, picks the lowest pinball loss among those that pass.

	Breaches	Kupiec p	Ind. p	CC p	Rolling red	Z2 zone	Pinball ×10⁴	Passes
Historical	13	0.376	0.009	0.022	0.00	yellow	3.56	no
Normal	25	0.000	0.002	0.000	0.40	yellow	3.25	no
Student-t	20	0.006	0.006	0.000	0.31	yellow	3.14	no
Cornish-Fisher	9	0.728	0.002	0.007	0.00	green	4.41	no
Monte Carlo	20	0.006	0.006	0.000	0.29	yellow	3.23	no
EVT	16	0.084	0.023	0.017	0.23	yellow	3.20	no
GARCH-FHS	12	0.555	0.591	0.726	0.00	green	2.13	yes
GJR-FHS	13	0.376	0.560	0.569	0.00	green	2.02	yes
Conditional EVT	14	0.241	0.530	0.411	0.00	green	2.04	yes
GJR conditional EVT	14	0.241	0.530	0.411	0.00	green	1.99	yes

Table 8. Selection on 2019–2022 at 99%, 1008 days, about 10 breaches expected.

Models that ignore current volatility fail on clustering, however good their total count. The rule picks **GJR conditional EVT**: GJR-GARCH volatility, where falls raise volatility more than rises, with a GPD tail on the residuals (McNeil and Frey, 2000). It had 14 breaches against 10.1 expected. Every breach has a documented market cause, and six were the first days of a new crisis. It has no red rolling window, although 34% of its windows are yellow.

	GFC 2008	COVID crash 2020	Rate shock 2022
Historical	16	8	5
Normal	26	9	11
Student-t	21	8	8
Cornish-Fisher	6	5	4
Monte Carlo	21	8	9
EVT	19	8	6
GARCH-FHS	6	4	2
GJR-FHS	5	4	2
Conditional EVT	5	4	2
GJR conditional EVT	4	4	2
Expected	3.0	0.2	2.0

Table 9. 99% VaR breaches inside each crisis window, 2008 to 2022, on the candidate portfolio.

Under stress the gap between models is wide. Models without a volatility forecast were overrun in 2008 and 2022, with up to nine times the expected breaches. Volatility-aware models stayed near expectation there. The COVID crash overran every model, because no daily model can react to a fall that sharp. That is why the fund has a crisis budget and stress tests as well as a daily risk number.

One-shot test on 2023–2026

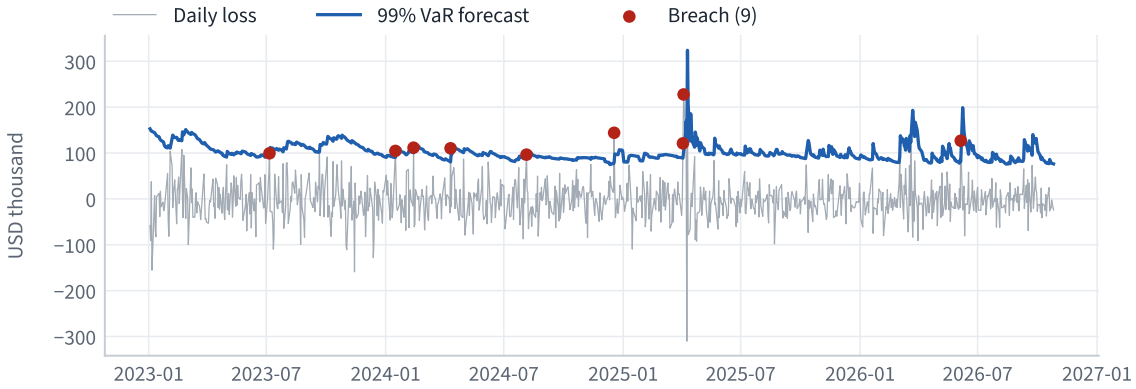


Figure 4. GJR conditional EVT on the one-shot test, 2023 to 26 October 2026.

The champion passed: 9 breaches against 9.6 expected over 955 days, Kupiec p = 0.86, conditional coverage p = 0.19, and the lowest pinball loss of eleven models (Table 10). Only three models passed; those without GJR's asymmetry failed because their breaches clustered. A separately written notebook reproduced the breaches and losses.

What the test does not show

About ten expected breaches give limited power, so the test shows the model is not detectably wrong, not that it is right. Over all of 2008–2026 the joint coverage test falls just below 5% because of clustering in 2008.

7. Champion–challenger and final test

A risk model should be challenged. We tested machine-learning challengers for 99% VaR under a rule fixed in advance: a challenger must be better on validation, pass coverage, and stay within 2% of the champion's loss on the final test. Comparisons used Diebold-Mariano tests with a Holm correction and a model confidence set.

LightGBM and an LSTM failed coverage on validation in every configuration and were dropped. Linear quantile regression stayed as the reported challenger: on validation it had 19 breaches (Kupiec p = 0.01), so it failed coverage, and its difference from the champion was not significant (Holm-adjusted p = 1.0). On the final test it passed coverage, but its loss was 5.4% worse than the champion's, so it was not adopted. No machine-learning model improved calibrated 99% forecasts for this portfolio, which is a useful result rather than a failed experiment.

One-shot final test, 2023-01-01 to 2026-10-26

	Breaches	Kupiec p	CC p	Passes	Pinball vs champion
Historical	6	0.215	0.0006	no	+10.8%
Normal	7	0.384	0.0017	no	+10.0%
Student-t	7	0.384	0.0017	no	+13.4%
Cornish-Fisher	5	0.103	0.0001	no	+30.9%
EVT	7	0.384	0.0017	no	+13.4%
Monte Carlo	7	0.384	0.0017	no	+14.9%
GARCH-FHS	8	0.604	0.0041	no	+4.2%
Conditional EVT	9	0.857	0.0077	no	+1.6%
GJR-FHS	7	0.384	0.0794	yes	+2.2%
GJR conditional EVT	9	0.857	0.1893	yes	+0.0%
Linear QR	10	0.885	0.2333	yes	+5.4%

Table 10. All eleven models on 955 common days, about 9.6 breaches expected.

8. Stress testing

Historical replays

We replay each crisis with today's target weights, rebalanced daily. This is the cautious convention, since it keeps the risky funds at weight while they fall.

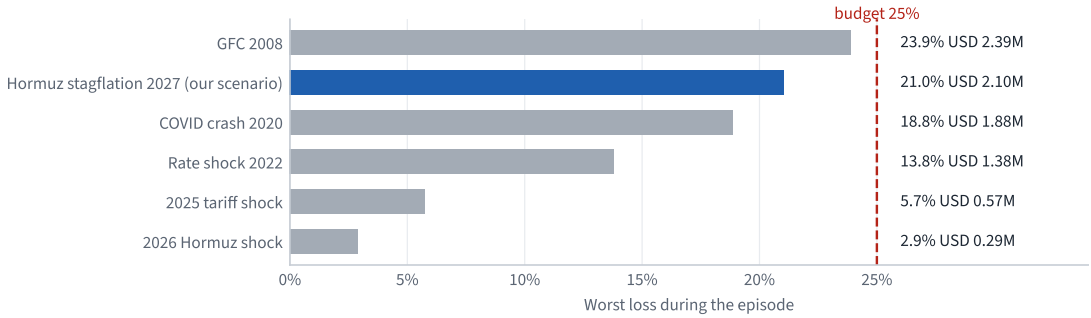


Figure 5. Peak-to-trough loss with today's target weights rebalanced daily, the cautious convention.

	Start	End
GFC 2008	2007-10-09	2009-03-09
COVID crash 2020	2020-02-19	2020-03-23
Rate shock 2022	2022-01-03	2022-10-12
2025 tariff shock (pre-registered)	2025-02-19	2025-04-08
2026 Hormuz shock (pre-registered)	2026-02-27	2026-03-30

Table 11. Replay windows, fixed in config.yaml. The two 2025–26 windows were written down before their results were seen.

- **GFC 2008:** 23.9%, the closest to the budget. Bonds helped.
- **COVID 2020:** 18.8%, deepened by the energy holding as oil collapsed.
- **Rate shock 2022:** 13.8%. Stocks and bonds fell together, so the bond hedge failed.
- **Recent episodes:** the 2025 tariff shock cost 5.7% and the 2026 Hormuz shock 2.9%, because energy gained.

Hypothetical scenario: Hormuz stagflation 2027

Our scenario is the crisis bonds cannot hedge. Over three quarters the 10-year yield rises 250 bp, oil 40%, VIX 25 points, the dollar 8% and credit spreads 150 bp, and AGG then has one panic day of −5%, like 10 October 2008 when bond-fund prices fell below the value of their bonds.

Fund moves come from factor betas estimated on stressed weeks. Each fund's log move is converted to a simple return before aggregation, because only simple returns add across assets. The scenario costs **21.0%**. It is the most severe candidate still within history, and the mapping reproduced 2022 at 17.8% predicted against 18.8% actual. A version beyond the historical record would cost about 25.3%, just over the budget, and we show it to the committee rather than leave it out.

The aggregation was corrected after the methodology freeze. This moved the scenario from 21.09% to 21.04% (USD 4,363), did not touch the test or the verdict, and is logged in the freeze record with both figures.

Diversification

	Funds one by one	Portfolio	Saved, USD
GFC 2008	28.43%	23.90%	452,769
COVID crash 2020	20.18%	18.85%	133,139
Rate shock 2022	18.56%	13.80%	475,854

Table 12. Crisis loss if each fund's loss is added up, against the portfolio's loss: the difference is what diversification saved.

Diversification saved between USD 133,000 and USD 476,000 in these crises. It helped least in COVID, when everything was sold for cash, which is the weakness a conservative investor should expect.

Reverse stress

Instead of asking what a crisis costs, we ask which crisis costs 25%. The answer is a broad equity crash with bonds not helping.

	SPY	EFA	AGG	XLE
In calm markets	-50.8%	-56.1%	-6.9%	-73.3%
In a crisis	-48.2%	-52.8%	-8.5%	-69.7%

Table 13. Most plausible fund moves that lose 25% in 21 trading days. A normal model calls this rarer than once in a million years, even in a crisis. Real crises are more frequent, so read the direction, not the odds.

Innovation and optional risks

We studied four extensions on the 35% candidate, using data to 2022.

- **Regimes:** a two-state model fitted to 2018 finds 99% ES 4.7 times higher in the high-volatility state, which lasts about 21 days against 79 for the calm state.
- **Dependence:** a t-copula ($\nu = 10.6$) raises 99% ES by 4.9% against a Gaussian copula; a joint SPY, EFA and EEM crash has probability 1.92% (Gaussian 1.72%, history 2.40%).
- **Macro:** the stock–bond correlation rose from 0.04 at end-2021 to 0.32 at end-2022; +100 bp on the 10-year yield costs about 2.2%.

	Return	Volatility	Max drawdown	99% ES	Calmar	Turnover
Static	4.91%	7.60%	21.13%	2.09%	0.23	0.12
Overlay to AGG	5.16%	6.89%	20.02%	1.86%	0.26	0.41
Overlay to T-bills	4.55%	6.25%	18.18%	1.62%	0.25	0.71

Table 14. Volatility-target overlay on the 35% frontier candidate, data to 2022. Studied, not adopted.

The volatility-target overlay into AGG raised return and cut drawdown on this data, at three times the turnover. We also tested an automatic de-risking trigger: it helped on 2008–2018, where it was designed, but made results worse on 2019–2022 for every setting. We therefore adopted neither. The committee meets when risk stays high, and nothing is sold automatically.

Crowd simulation

We also gave the same event to a MiroFish agent simulation, in which AI investors react and describe the market moves. Their answers set only direction and strength; the sizes come from history over the same three quarters, and the moves pass through the same stress test. Three runs gave losses of 14.5–19.4%, an average of 17.6%, against 21.0% for our scenario. This is a comparison only: it is not tested against real events and is not used in the recommendation.

Currency

The fund measures everything in US dollars but pays costs in Sri Lanka. We analysed LKR exposure and report it as a limitation, not a position, since the brief's universe has no rupee asset.

9. Risk budget and recommendation

Position sizing

Position sizing uses each holding's marginal contribution to ES. Euler's theorem splits the total, so the contributions add up exactly.

$$\mathrm{ES}_i = w_i \times \mathrm{E}[\, -R_i \mid L \geq \mathrm{VaR}\,] \text{ and } \Sigma\, \mathrm{ES}_i = \mathrm{ES}$$

	Weight	Share of 99% ES	ES share ÷ weight
SPY	15.57%	33.15%	2.13
XLE	10.00%	29.90%	2.99
EFA	9.43%	21.43%	2.27
AGG	65.00%	15.52%	0.24

Table 15. Euler contributions of each position to 99% ES, 2004 to the decision date. They add up to 100%.

No fund carries more than 40% of the tail risk, so the sizing rule moved nothing. The largest share is SPY at 33%. XLE is the most risk-dense holding, at three times its weight. AGG is 65% of the money but 16% of the tail risk, which is how a conservative portfolio should look.

Verdict

The rule has three outcomes. ACCEPT if a portfolio meets the objective inside every limit. ADJUST THE RISK LIMIT if one would with the crisis budget loosened by at most 5 points. REJECT otherwise. We recorded in advance that the verdict would turn on real yields: on end-2022 yields the same rule gave REJECT, and a 10-year TIPS yield of about 2.58% would flip it. On the decision date it was 2.66%, and the rule gave **ACCEPT at 35% equity**.

How close it is

ACCEPT is lost if the real yield falls by 0.15 points or if the equity premium is below 4.15%, against our 4.5%. Across 18 assumption combinations it is ACCEPT in 7, ADJUST in 4 and REJECT in 7. The largest single swing is tax: if AGG's interest is taxed at 15%, the base case becomes REJECT.

	exempt, +30 bp	exempt, +50 bp	exempt, +90 bp	taxed, +30 bp	taxed, +50 bp	taxed, +90 bp
ERP 3.5%	REJECT -29	REJECT -16	ACCEPT +10	REJECT -70	REJECT -57	REJECT -31
ERP 4.5%	ADJUST -2	ACCEPT +11	ACCEPT +37	REJECT -43	REJECT -30	ADJUST -4
ERP 5.5%	ACCEPT +25	ACCEPT +38	ACCEPT +64	ADJUST -15	ADJUST -2	ACCEPT +24

Table 16. Verdict and margin (bp a year) over the assumption grid. Columns: AGG interest tax-exempt or taxed at 15%, and AGG's spread over Treasuries.

Over ten years, a block bootstrap puts the chance of meeting the objective at about 49%, with a median real return of 3.5% a year and an 8% chance that real value ends below 80% of today's. These figures are context, not part of the rule.

Why this suits a conservative institution

- **It protects capital first.** Each limit holds, and the worst crisis on record stays inside the budget.
- **Income pays the scholarships.** Dividends and interest of about USD 327,000 a year (3.27%) cover the 3% spending rate, so the fund does not need to sell in a crisis.
- **It is simple and liquid.** Four liquid ETFs and a quarterly rebalance.
- **Its weaknesses are known.** The committee sees in advance where the bond hedge fails and what would change the answer.

Governance

- Rebalance to target weights every quarter.
- If the daily risk forecast stays above the limit for 5 days, the committee meets. Nothing is sold automatically, because an automatic de-risking rule made results worse out of sample.
- Review the spending rate, the limits and the return assumptions every year.
- Obtain tax advice on AGG's interest before investing.

10. Assumptions and limitations

The table lists each assumption, why it matters and what we do about it. Three matter most:

- **The return objective** rests on an assumed equity premium and on AGG's tax treatment.
- **Models cannot price a crisis unlike any in the data**, and the sample has no 1970s-style inflation.
- **The fund measures everything in US dollars** while paying scholarships in rupees.

Data

Assumption	Why it matters	Mitigation
History is representative of the future	Models fitted on 2004–2018 and chosen on 2019–2022 may miss new regimes	Stress tests, EVT and a one-shot test on 2023–2026
One frozen snapshot (to 2026-10-26)	Numbers change with new data	Hashed manifest; refresh is explicit and logged
ETF prices proxy the investable portfolio	Tracking error and fees are inside the ETF prices	Expense ratios are not deducted twice
FRED keeps only 3 years of the ICE high-yield spread	Credit stress needs long history	Moody's Baa–Treasury spread (BAA10Y) used instead

Models

Assumption	Why it matters	Mitigation
Volatility is time-varying	Unconditional models lag in crises	GJR conditional EVT (the tested model) reacts to today's volatility
GARCH parameters are uncertain	Recent fits sit near the persistence boundary ($\alpha + \beta \approx 1$), which inflates multi-day forecasts	Disclosed; 10-day figures shown with historical and $\sqrt{h}$ comparisons
EVT rests on few extreme points	Wide confidence intervals at 99.9%	Profile and bootstrap intervals; two threshold rules
Few tail observations to validate 99% VaR	About 10 exceptions expected in 4 years; low test power	Confidence intervals, power figures and 95% results
Exceptions cluster in 2008–2018	Independence is rejected on that period; over 2008–2026 the joint test fails at 5%	Disclosed on the Risk page; selection on 2019–2022, test on 2023–2026
Normal and Cornish-Fisher are benchmarks only	The first understates fat tails; the second is outside its valid range for tails this fat	Never used for decisions; Cornish-Fisher carries a warning

Portfolio

Assumption	Why it matters	Mitigation
Expected returns are estimates (today's yields plus a 4.5% equity premium)	The verdict turns on them	18-case robustness grid and the what-if on the first page
The bond fund's interest is assumed free of US withholding tax	If it were taxed like dividends, the verdict would be REJECT	A tax adviser must confirm before money is invested
Costs approximated at 5 bps per side	No market impact in a crisis	Quarterly rebalancing keeps turnover low
Fully invested (cash only as a sensitivity)	No liquidity buffer by default	T-bill sleeve sensitivity case

Stress

Assumption	Why it matters	Mitigation
Historical paths may not repeat	The next crisis may differ	Factor-driven hypothetical and reverse stress
Factor betas are unstable across regimes	Scenario impacts are estimates	Stress-week betas, SE bands
The hypothetical scenario is one of many	Shock sizes and horizon are judgement	Anchored on historical extremes; sizes, horizon and credit measure fixed before the freeze
The scenario treats the fund as unrebalanced during the episode	Each fund's scenario move is applied to its starting weight	Corrected after the freeze so that fund moves are added as simple returns (0.04 points, logged)
The bad-day risk budget does not see the stagflation scenario	Bonds carry little bad-day risk but most of the scenario loss	The crisis budget is checked separately, with the scenario included

Decision

Assumption	Why it matters	Mitigation
A narrow margin over the objective	A small change in assumptions flips the verdict	Shown on the first page; review every year
The 10-year odds of meeting the objective are about even	Spending may need to fall after bad years	Spending is smoothed over 12 quarters; annual review
USD base while spending is in LKR	FX risk is outside the model	Disclosed; LKR view on the innovation page
No derivatives, margin or ESG screens	Out of scope for the universe	Stated in the IPS

11. References

Acerbi, C. and Székely, B. (2014). Back-testing expected shortfall. Risk.

Acerbi, C. and Tasche, D. (2002). On the coherence of expected shortfall. Journal of Banking & Finance 26.

Artzner, P., Delbaen, F., Eber, J.-M. and Heath, D. (1999). Coherent measures of risk. Mathematical Finance 9.

Balkema, A. and de Haan, L. (1974). Residual life time at great age. Annals of Probability 2.

Barone-Adesi, G., Giannopoulos, K. and Vosper, L. (1999). VaR without correlations. Journal of Futures Markets 19.

Basel Committee on Banking Supervision (1996). Supervisory framework for the use of backtesting.

Christoffersen, P. (1998). Evaluating interval forecasts. International Economic Review 39.

Diebold, F. and Mariano, R. (1995). Comparing predictive accuracy. Journal of Business & Economic Statistics 13.

Fissler, T. and Ziegel, J. (2016). Higher order elicitability and Osband's principle. Annals of Statistics 44.

Glosten, L., Jagannathan, R. and Runkle, D. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. Journal of Finance 48.

Hansen, P., Lunde, A. and Nason, J. (2011). The model confidence set. Econometrica 79.

Koenker, R. and Bassett, G. (1978). Regression quantiles. Econometrica 46.

Kupiec, P. (1995). Techniques for verifying the accuracy of risk measurement models. Journal of Derivatives 3.

Ledoit, O. and Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. Journal of Multivariate Analysis 88.

Markowitz, H. (1952). Portfolio selection. Journal of Finance 7.

McNeil, A. and Frey, R. (2000). Estimation of tail-related risk measures for heteroscedastic financial time series. Journal of Empirical Finance 7.

Pickands, J. (1975). Statistical inference using extreme order statistics. Annals of Statistics 3.

Sharpe, W. (1964). Capital asset prices. Journal of Finance 19.

12. AI disclosure

AI Usage Disclosure

AI Tool Used: Claude (Anthropic), through Claude Code.

Purpose of AI Usage: AI was used as a productivity tool to accelerate development, support research, assist with debugging, improve documentation, and streamline repetitive tasks. The project was developed through the team's own research, engineering knowledge, critical thinking, coding, and collaborative problem-solving.

Research and Planning: The team conducted the financial and technical research, studied relevant methodologies, evaluated alternative approaches, and established the overall solution strategy. AI assisted with finding information, organizing references, and improving research efficiency.

System Architecture and Development: The team designed the architecture, defined system workflows, developed algorithms, wrote and reviewed code, and integrated the application components. Claude Code helped accelerate coding, generate boilerplate, troubleshoot errors, and refine implementations under the team's direction.

Financial Modeling and Analysis: The team selected financial methodologies, formulated investment assumptions, established risk constraints, implemented analytical calculations, evaluated portfolio strategies, and interpreted results. AI assisted with computational tasks and development efficiency, while financial reasoning and decisions remained with the team.

Testing and Validation: The team developed and reviewed tests, performed debugging, verified calculations, and evaluated system correctness. AI helped accelerate test preparation and issue identification. Results were checked using manual calculations and reconciliation between production code and analytical notebooks across 957 backtesting days.

Documentation and Reporting: The team developed the technical explanations, analytical findings, investment recommendations, and report content. AI assisted with drafting, language refinement, formatting, and organizing documentation.

Team Ownership and Accountability: The team was actively responsible for research, design, implementation, testing, evaluation, and final decision-making throughout the project. AI served as a supporting development tool to improve productivity and efficiency, not as a replacement for the team's technical or intellectual contributions.The team accepts full responsibility for the final submission.